

## Education

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|-----------|------------------------------------------------------------------------------------|--------------------------|
| Exp. 2022 | <b>Bachelor of Technology</b> , Dept. of Electrical Engineering, IIT Kanpur, India | <i>CPI: 9.82/10</i>      |
| May 2018  | <b>Grade XII (AISSCE)</b> , Mansukhbhai Kothari National School, Pune              | <i>Percentage: 96.6%</i> |
| May 2016  | <b>Grade X (AISSE)</b> , Mansukhbhai Kothari National School, Pune                 | <i>CGPA: 10/10</i>       |

## Achievements

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|----------|----------------------------------------------------------------------------------------------|------------------------|
| Dec 2019 | <b>Gold Medal</b> , Inter IIT Tech Meet 8.0 - DRDO SASE's UAV Fleet Challenge                | <i>IIT Roorkee</i>     |
| Apr 2019 | <b>Academic Excellence Award</b> , for exceptional performance in the academic year 2018-19  | <i>IIT Kanpur</i>      |
| Jun 2018 | <b>AIR 408</b> , Joint Entrance Examination - Advanced, among 230,000 candidates             | <i>India</i>           |
| May 2018 | <b>AIR 513</b> , Joint Entrance Examination - Mains, among 1.1 million candidates            | <i>India</i>           |
| Mar 2017 | <b>AIR 214</b> , Kishore Vaigyanik Protsahan Yojana (KVPY) - SA, conducted by IISc Bangalore | <i>IISER Pune</i>      |
| Nov 2016 | <b>State Rank 15</b> , Regional Mathematical Olympiad                                        | <i>Maharashtra-Goa</i> |

## Projects

### DRDO SASE's UAV Fleet Challenge

8TH INTER IIT TECH MEET, IIT ROORKEE

AerialRobotics-IITK/inter\_iit\_uav\_fleet

Nov-Dec. 2019: [interiittech.org](http://interiittech.org)

- Won the **sole Gold Medal** amongst **20** participating IITs, achieving a **perfect score** (400/400) and a special mention
- Developed a **fleet of MAVs** capable of **collaboratively surveying** a 1600 sq.m. grass field to locate green boxes
- Implemented a finite-state machine for **high-level planning** of the complete survey mission flow using Boost MSM
- Extracted the **target GPS coordinates** by implementing HSV-thresholded **box-contour detection** using OpenCV and subsequent **pose transformations** based on the pinhole camera model, achieving **centimeter-level precision**
- Setup a multimaster architecture for **intra-fleet communication** to accurately track number of detected targets

### International Micro Aerial Vehicle Challenge 2019, Madrid, Spain

OUTDOOR PROBLEM STATEMENT — TEAM AERIAL ROBOTICS IITK

AerialRobotics-IITK/imav2019

Mar-Oct. 2019: [imav2019.org](http://imav2019.org)

- Amongst the **top 15 student teams internationally** to qualify for the competition, *only* student team from India
- Researched optimal mission planning for **coordinated surveying** of a 30000 sq.m. area using a **fleet of 3 MAVs**
- Implemented a **floodfill-based vision algorithm** for the detection of multiple and differently coloured mailboxes
- Applied **visual servoing** on the detection feedback for the delivery of packages into the detected mailboxes
- Engineered a **publish-subscribe messaging system** for intrafleet sharing of delivery status and drop locations
- Designed the **state transition diagram** for the complete mission plan and implemented it within an FSM
- Explored the use of **feature detection algorithms** for the detection of a **crashed MAV** within the surveying region
- Implemented a **CNN framework** for crashed MAV detection task based on **YOLOv3** using drone image datasets

### PETcat: Project Lead

LONG TERM PROJECT — Science and Technology Council, IIT Kanpur

RoboticsClubIITK/PETcat

Apr. 2020 - Present

- Driving the development of an **innovative, fully autonomous** and **highly interactive robotic cat**
- Managing a team of 25 students working towards building the **entire software stack** enabling the cat's autonomy
- Responsible for the **implementation of the dynamics** of the CAD model in **simulation** for rigorous software testing
- Mentoring multiple subteams in building the **vision, voice, localization and avoidance modules** for the cat

## Planning Library

SELF PROJECT — *Aerial Robotics IITK*

AerialRobotics-IITK/ariitk\_planning\_library

Apr-May, 2020

- Studied the generation of **Truncated and Euclidean Signed Distance Fields** from depth pointclouds using Voxelbox
- Experimented with the **generation of sparse 3D graphs** by skeletonizing the Voronoi diagram of the ESDF
- Implemented a **global planner using A\*** on the sparse graph for **real-time path planning** in a known environment
- Built a **local planner for inter-waypoint planning** by using A\* on a graph generated by randomly sampling points
- Explored frontier-based navigation for **autonomous exploration of unknown environments** using TSDFs

## Autonomous Navigation in Rough Terrain Environments

SUMMER PROJECT MENTOR — *Robotics Club IITK*

RoboticsClubIITK/2020\_rover

May-Jul, 2020

- Mentored a team of 15 students in implementing a complete **navigation stack** on a **six-wheeled rover** in simulation
- Built the **simulation environment** for the project in Gazebo using open-source terrain models and rover URDFs
- Configured servo motors, a 2D **LiDAR**, **depth camera** and a GPS sensor via the use of **ROS controllers** and plugins
- Guided the implementation of GPS and **position controllers**, **fiducial marker detection**, LiDAR-based **obstacle avoidance** and **path planning** for the rover

## Basics of Reverse Engineering the Brain

SUMMER WORKSHOP, BRAIN AND COGNITIVE SOCIETY

ashwin2802/BCS\_Workshop\_Apr\_20

Apr, 2020

- Evaluated various **CIFAR10 image classification architectures** and implemented an **RL agent using Q-learning**
- Studied and implemented Atkinson-Shiffrin's **Multi-Store Model of Memory** and verified the **serial position effect**
- Verified the postulates of Feature Integration Theory's **Model of Attention** by modelling feature search in images
- Attempted to **replicate the results of research** on *the role of categorization in visual search for orientation* by **conducting a survey** generated using Psytoolkit and statistically analysing the data from 10 participants

## Simulation of a Beam Splitter

COURSE PROJECT, EE683 — *Supervisor: Prof. Shilpi Gupta*

ashwin2802/EE683

Oct-Nov, 2019

- Modelled **single photon sources** in Python with uniform, exponential and logarithmic emission distributions
- Derived expressions for the **beam splitter operator** applying basic principles of quantum mechanics and optics
- Implemented the derived expressions to **verify experimental results and postulates** such as the *Hong-Ou-Mandel effect* by simulating various input configurations and computing the **second-order output correlation**

## Bar Pantograph

COURSE PROJECT, TA201 — *Supervisor: Prof. Anish Upadhyaya*

Aug-Nov, 2019

- Designed a simple parallelogram bar mechanism **prototype that could copy designs** drawn by the operator
- Demonstrated proof-of-concept via **AutoCAD assembly** and built the prototype working in a *team of 5*
- Acquired **hands-on experience** of basic manufacturing processes like *welding, brazing, casting and forging*

## Line Follower Robot

SUMMER PROJECT, ROBOTICS CLUB IITK

ashwin2802/FLF

May-Jun, 2019

- Won the **Best Documentation Award** amongst approximately **40+** student projects in SnT Summer Camp '19
- Designed and **manufactured** a line follower robot equipped with an *Arduino Uno, IR sensor arrays and microservos*
- Implemented a **PID controller** based upon the feedback from the sensors for **precise line following**
- Developed a **gradient-descent based tuning algorithm** to calculate the PID gains in an initial calibration phase
- Implemented a simple **left-hand rule algorithm** for handling branches and dead ends in the path to be followed
- Researched the use of **graph-based path-finding algorithms** for handling branches and dead-ends in the course

## Pong: An AI Based Approach

ADVANCED TRACK, ESC101 — Supervisor: Prof. Swaprava Nath

ashwin2802/Pong

Feb-Apr. 2019

- Designed and **built a Pong application** with 2-player as well as 1-player interfaces using *Thorpy* and *Pygame*
- Simulated **difficulty levels** for the 1-player interface's AI by implementing **various decision methods** modularly
- Implemented simple algorithmic decision strategies like **random movement** and **ball tracking**
- Experimented with a **deep-learning based model** by using *pillow* to input the game state into a *CNN* to decide the action and then trained the model using **reinforcement methods** with different reward strategies

## Positions of Responsibility

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### Team Head — Aerial Robotics IITK

Apr. 2020 - Present

- Responsible for **developing** and **maintaining** the **software stack** for our **custom-built fleet** of aerial robots
- Participating in **international aerial robotics competitions**, IMAV 2019 & IARC 2021 leading a team of **15** members
- Mentored introductory projects on **autonomous FPV racing** and **landing on moving platforms** for new members
- Established **proper guidelines and standards** for software **development, documentation and licensing**

### Manager, Campus Wing — Outreach Cell

Apr. 2020 - Present

- Part of a **3-tier** team working on **building student-alumni relations** and **publicizing achievements** of IITK students
- Managing a team of **7** secretaries in various tasks such as **content creation** for the **#ThatsIITK** Quora Blog, execution of the **Distinguished Alumni Interview Series**, creation of **alumni databases** and supporting the DoRA office
- Brainstormed and conducted **Legacy Hunt**, an online event to **create awareness** about notable IITK alumni

### Secretary & Summer Project Mentor — Robotics Club

Apr. 2019 - Jul. 2020

- Accountable for **conducting competitions**, organizing and delivering **robotics-related lectures and workshops**
- Supervised approximately **50** freshers to **win all the robotics events** in Takneek '19 - with the **Robotics Trophy**
- Developed curriculum to teach the **basics of ROS** to approximately **30** participants in the club's 2019 Winter Camp
- Mentored **15** first-year students in implementing a **complete navigation stack** on a **six-wheeled rover**

### Student Guide & Academic Mentor — Counselling Service

Apr. 2019 - Apr. 2020

- Part of the **organizing** team for the **7-day Orientation programme** for the incoming Batch of 2019
- Helped a group of **6** first year students **acclimatize** to the campus environment, **resolving any problems** they faced
- Conducted **remedial lectures** and **discussion sessions** for first year students in the courses **MATHEMATICS-I/II**
- Provided **personal mentoring** to academically **underperforming students** and helped them overcome difficulties

## Skills

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**Programming Languages** C/C++, Python, MATLAB, Java, R, MySQL, Javascript

**Robotics** ROS, OpenCV, PX4, Gazebo, RViz, Arduino

**Utilities**  $\text{\LaTeX}$ , Markdown, Git, Autodesk Inventor, LTSpice

**Languages** English, Hindi, Marathi, Konkani, Kannada

## Relevant Coursework

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**Computer Science** Fundamentals of Computing, Data Structures and Algorithms

**Electrical Engineering** Microelectronics-I, Signals, Systems and Networks, Introduction to Electrical Engineering, Control Systems Analysis, Quantum and Wave Phenomena

**Mathematics** Probability and Statistics, Linear Algebra, Complex Analysis, Partial Differential Equations

**Coursera** Deep Learning Specialization, Tensorflow in Practice Specialization (*Online Courses*)